

Fig. 1: Block diagram of induction lamp system

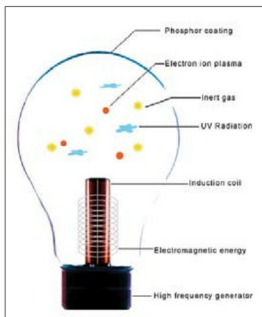


Fig. 2: Self-ballasted cavity induction lamp

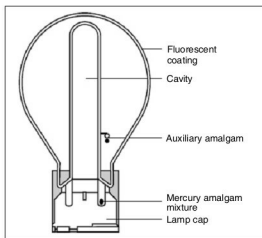


Fig. 3: The discharge vessel

phosphor coating on the inside surface of the glass globe, making it glow with visible light.

The basic idea is similar to fluorescent lamps, except they pass an electric

current through the gas to generate ultraviolet light. Elimination of electrodes extends the bulb life as electrodes are the limiting factor in bulb life as gas reacts with metal electrodes in normal lamps.

Components of the system

Fig. 1 shows the block diagram of the induction lamp system. The system comprises three components—the high-frequency generator, the power coupler and the lamp. The power coupler transfers energy from the high-frequency (HF) generator to the discharge inside the glass bulb using an antenna that contains an induction coil. The power coupler also has a heat-conducting rod with mounting flange, which allows the induction lamp system to be mechanically attached to the luminaire.

The HF generator consists of an oscillator, which supplies the high-frequency power to the antenna to initiate and maintain a gas discharge in the glass vessel. A pre-conditioning circuit (EMI filter and power factor correction) takes care of the oscillator power supply and filtering.

The induced current circulates through the metal vapour or gas, causing the acceleration of free electrons, which collide with the atoms and bring electrons to a higher-level energy state. Electrons from excited atoms fall back from this higher energy

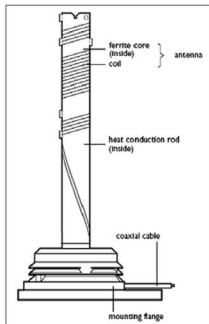


Fig. 4: Power coupler

state to the lower stable level and consequently emit UV photons. The UV photons interact with the fluorescent powder on the wall of the discharge vessel, whereby visible light is generated.

Discharge vessel. Like fluorescent lamps, induction lamps have a discharge space called discharge vessel, which is basically a glass bulb containing a low-pressure mercury vapour (see Fig. 3). This vessel consists of a tubular re-entrant portion where the non-contact antenna can be inserted. The re-entrant is coated with an electron-emissive material, while the inside of the bulb is coated with a triphosphor fluorescent powder and anti-blackening agents. The inside of the bulb has a conductive fluorine-tin-oxide coating to screen undesirable radiation and is capacitively coupled to an external silver coating on the rear. A mercury amalgam mixture, housed in a reservoir attached to the discharge vessel, helps maintain the light output at a more or less constant level over a large temperature range. To ensure a short run-up time, an auxiliary amalgam is positioned close to the discharge and a glass bulb is fixed to the bottom of the power coupler.